

TP 4 – Mise en place d'un DNS

karadag Nissa

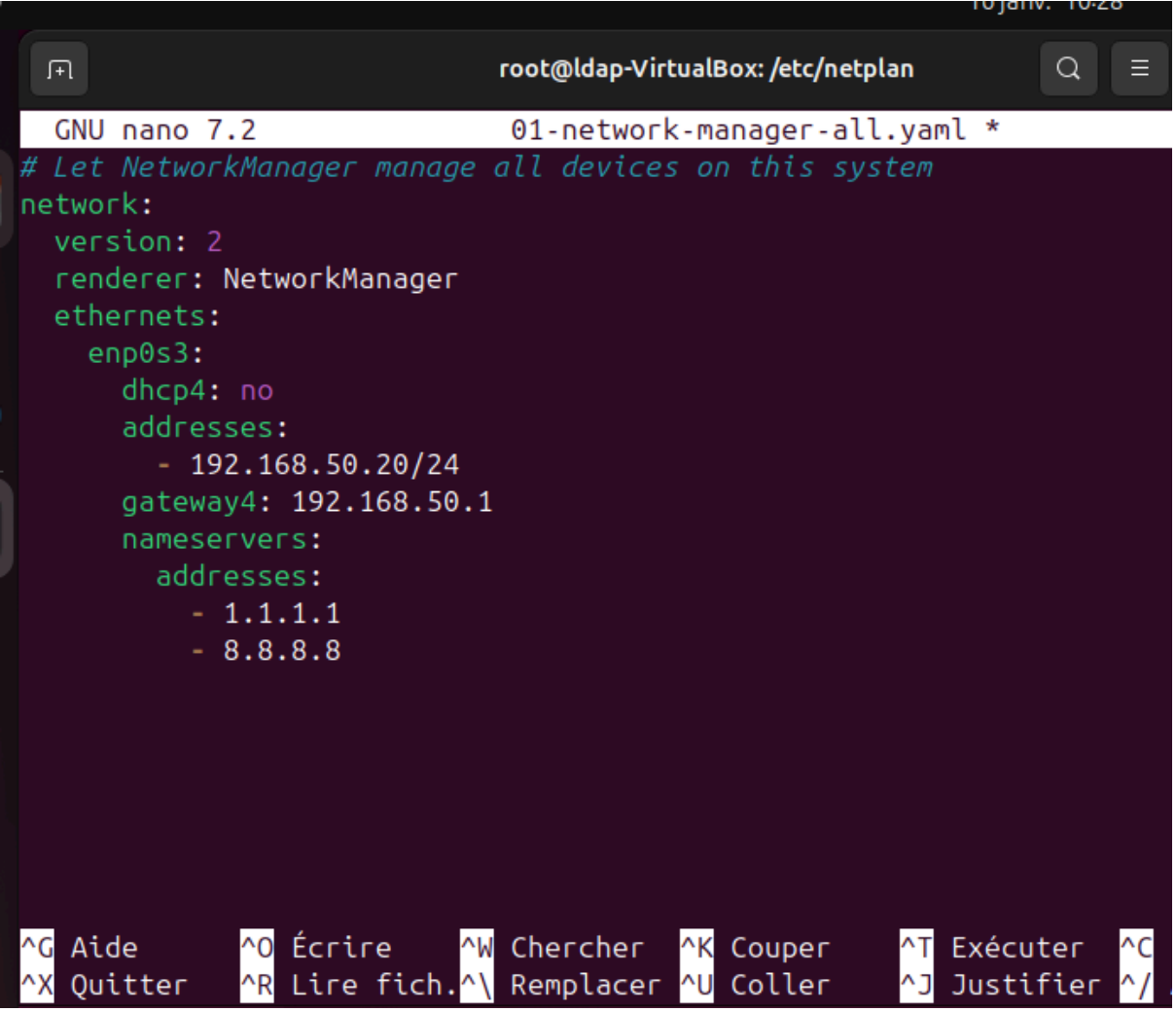
srv ldap = id & mdp : ldap

srv ntp = id & mdp : ntp

on mets des ip static:

- sudo nano /etc/netplan/01-network-manager-all.yaml

dans le fichier config on mets:



```
root@ldap-VirtualBox: /etc/netplan
GNU nano 7.2 01-network-manager-all.yaml *
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.50.20/24
      gateway4: 192.168.50.1
      nameservers:
        addresses:
          - 1.1.1.1
          - 8.8.8.8

^G Aide      ^O Écrire    ^W Chercher  ^K Couper    ^T Exécuter  ^C
^X Quitter   ^R Lire fich.^_ Remplacer  ^U Coller    ^J Justifier ^/ .
```

- sudo chmod 600 /etc/netplan/*.yaml
- sudo netplan apply

on verifie:

- ip a
- ip route

```
root@ldap-VirtualBox:/etc/netplan# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b5:c3:b2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.20/24 brd 192.168.50.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:feb5:c3b2/64 scope link
        valid_lft forever preferred_lft forever
root@ldap-VirtualBox:/etc/netplan# ip route
default via 192.168.50.1 dev enp0s3 proto static metric 20100
192.168.50.0/24 dev enp0s3 proto kernel scope link src 192.168.50.20 metric 100
root@ldap-VirtualBox:/etc/netplan#
```

Le serveur SRV-LDAP est configuré avec une adresse IP statique (192.168.50.20) et une passerelle définie à 192.168.50.1, correspondant au routeur du réseau interne, afin d'assurer le routage des communications entre les différentes machines de l'infrastructure.

on installe le serveur dns:

- sudo apt update
- sudo apt install bind9 bind9utils dnsutils -y

```
[sudo] mot de passe de ldap :
root@ldap-VirtualBox:/home/ldap# apt update
Atteint :1 http://archive.ubuntu.com/ubuntu noble InRelease
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Réception de :3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Réception de :4 http://archive.ubuntu.com/ubuntu noble/main Translation-fr [491 kB]
Réception de :5 http://archive.ubuntu.com/ubuntu noble/restricted Translation-fr [3 292 B]
Réception de :6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15,0 MB]
Réception de :7 http://archive.ubuntu.com/ubuntu noble/universe Translation-fr [3 898 kB]
Réception de :8 http://archive.ubuntu.com/ubuntu noble/universe Translation-en [5 982 kB]
Réception de :9 http://archive.ubuntu.com/ubuntu noble/universe amd64 Components
```

```

root@ldap-VirtualBox:/home/ldap# apt install bind9 bind9utils dnsutils -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  bind9-dnsutils bind9-host bind9-libs bind9-utils
Paquets suggérés :
  bind-doc
Les NOUVEAUX paquets suivants seront installés :
  bind9 bind9-utils bind9utils dnsutils
Les paquets suivants seront mis à jour :
  bind9-dnsutils bind9-host bind9-libs
3 mis à jour, 4 nouvellement installés, 0 à enlever et 287 non mis à jour.
Il est nécessaire de prendre 1 887 ko dans les archives.
Après cette opération, 1 688 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-
host amd64 1:9.18.39-0ubuntu0.24.04.2 [50,5 kB]
Réception de :2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-

```

on vérifie:

- systemctl status bind9

```

root@ldap-VirtualBox:/home/ldap# systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: en>
   Active: active (running) since Fri 2026-01-16 10:36:16 CET; 47s ago
     Docs: man:named(8)
  Main PID: 3773 (named)
    Status: "running"
     Tasks: 8 (limit: 4597)
    Memory: 23.0M (peak: 23.4M)
       CPU: 113ms
    CGroup: /system.slice/named.service
           └─3773 /usr/sbin/named -f -u bind

janv. 16 10:36:16 ldap-VirtualBox named[3773]: managed-keys-zone: loaded serial>
janv. 16 10:36:16 ldap-VirtualBox named[3773]: zone 0.in-addr.arpa/IN: loaded s>
janv. 16 10:36:16 ldap-VirtualBox named[3773]: zone 127.in-addr.arpa/IN: loaded>
janv. 16 10:36:16 ldap-VirtualBox named[3773]: zone localhost/IN: loaded serial>
janv. 16 10:36:16 ldap-VirtualBox named[3773]: zone 255.in-addr.arpa/IN: loaded>
janv. 16 10:36:16 ldap-VirtualBox named[3773]: all zones loaded
janv. 16 10:36:16 ldap-VirtualBox named[3773]: running
janv. 16 10:36:16 ldap-VirtualBox systemd[1]: Started named.service - BIND Doma>
janv. 16 10:36:26 ldap-VirtualBox named[3773]: managed-keys-zone: Unable to fet>
janv. 16 10:36:26 ldap-VirtualBox named[3773]: resolver priming query complete:>
lines 1-22/22 (END)

```

on configure:

- sudo nano /etc/bind/named.conf.options

```
GNU nano 7.2 /etc/bind/named.conf.options *
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk.  See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.
    recursion yes;
    allow-recursion { 192.168.50.0/24 };
    listen-on { 127.0.0.1; 192.168.50.20 };
    forwarders { };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys.  See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on-v6 { any; };
};

^G Aide      ^O Écrire    ^W Chercher  ^K Couper    ^T Exécuter  ^C Emplacement
^X Quitter   ^R Lire fich.^_ Remplacer  ^U Coller    ^J Justifier ^/ Aller ligne
```

(192.168.50.20 = IP de SRV-LDAP)

on declare les zones DNS:

- sudo nano /etc/bind/named.conf.local

```
root@ldap-VirtualBox: /home/ldap
GNU nano 7.2 /etc/bind/named.conf.local *
zone "zl-cyber.local" {
    type master;
    file "/etc/bind/db.zl-cyber.local";
};

zone "50.168.192.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192.168.50";
};

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

^G Aide      ^O Écrire    ^W Chercher  ^K Couper    ^T Exécuter  ^C Emplacement
^X Quitter   ^R Lire fich.^_ Remplacer  ^U Coller    ^J Justifier ^/ Aller ligne

root@ldap-VirtualBox: /home/ldap
GNU nano 7.2 /etc/bind/named.conf.options *
options {
    directory "/var/cache/bind";

    recursion yes;
    allow-recursion { 192.168.50.0/24; };

    listen-on { 127.0.0.1; 192.168.50.20; };

    forwarders { };

    dnssec-validation auto;

    listen-on-v6 { any; };
};
```

créer la zone:

- `sudo cp /etc/bind/db.local /etc/bind/db.zl-cyber.local`
- `sudo nano /etc/bind/db.zl-cyber.local`

```
root@ldap-VirtualBox:/home/ldap# nano /etc/bind/named.conf.options
root@ldap-VirtualBox:/home/ldap# nano /etc/bind/named.conf.local
root@ldap-VirtualBox:/home/ldap# sudo cp /etc/bind/db.local /etc/bind/db.zl-cyber.local
root@ldap-VirtualBox:/home/ldap# nano /etc/bind/db.zl-cyber.local
root@ldap-VirtualBox:/home/ldap#
```

```
root@ldap-VirtualBox:/home/ldap
GNU nano 7.2 /etc/bind/db.zl-cyber.local *
$TTL      604800
@          IN      SOA      srv-ldap.zl-cyber.local. admin.zl-cyber.local. (
                                2026011601      ; Serial
                                604800           ; Refresh
                                86400            ; Retry
                                2419200          ; Expire
                                604800 )         ; Negative Cache TTL
;
@          IN      NS       srv-ldap.zl-cyber.local.
srv-ldap   IN      A        192.168.50.20
```

on modifie aussi ici:

- sudo nano /etc/bind/db.192.168.50

```
root@ldap-VirtualBox:/home/ldap
GNU nano 7.2 /etc/bind/db.192.168.50 *
$TTL      604800
@          IN      SOA      srv-ldap.zl-cyber.local admin.zl-cyber.local. (
                                2026011601      ; Serial
                                604800           ; Refresh
                                86400            ; Retry
                                2419200          ; Expire
                                604800 )         ; Negative Cache TTL
;
@          IN      NS       srv-ldap.zl-cyber.local.
20         IN      PTR      srv-ldap.zl-cyber.local.
```

on exécute:

- sudo named-checkconf
- sudo named-checkzone zl-cyber.local /etc/bind/db.zl-cyber.local
- sudo named-checkzone 50.168.192.in-addr.arpa /etc/bind/db.192.168.50
- sudo systemctl restart bind9

```

root@ldap-VirtualBox:/home/ldap# named-checkconf
root@ldap-VirtualBox:/home/ldap# named-checkzone zl-cyber.local /etc/bind/db.zl-cyber.local
zone zl-cyber.local/IN: loaded serial 2026011601
OK
root@ldap-VirtualBox:/home/ldap# named-checkzone 50.168.192.in-addr.arpa /etc/bind/db.192.168.50
zone 50.168.192.in-addr.arpa/IN: loaded serial 2026011601
OK
root@ldap-VirtualBox:/home/ldap#

```

tout est ok donc:

on vérifie après le restart:

- dig @127.0.0.1 -x 192.168.50.20

```

root@ldap-VirtualBox:/home/ldap# systemctl restart bind9
root@ldap-VirtualBox:/home/ldap# dig @127.0.0.1 -x 192.168.50.20

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @127.0.0.1 -x 192.168.50.20
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12633
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 74becfadd72ec6a01000000696a115a8840ad65560f6983 (good)
;; QUESTION SECTION:
;20.50.168.192.in-addr.arpa.      IN      PTR

;; ANSWER SECTION:
20.50.168.192.in-addr.arpa. 604800 IN      PTR      srv-ldap.zl-cyber.local.

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Jan 16 11:22:18 CET 2026
;; MSG SIZE rcvd: 120

root@ldap-VirtualBox:/home/ldap#

```

La résolution inverse a été validée à l'aide de la commande dig -x 192.168.50.20, confirmant que l'adresse IP du serveur DNS principal correspond bien au nom [srv-ldap.zl-cyber.local](#).

serveur ntp:

on configure l'ip du réseau en statique

- `sudo nano /etc/netplan/01-network-manager-all.yaml`

```
GNU nano 7.2 /etc/netplan/01-network-manager-all.yaml *
# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: no
      addresses:
        - 192.168.50.21/24

    enp0s8:
      dhcp4: yes
```

- `sudo chmod 600 /etc/netplan/*.yaml`
- `sudo netplan apply`

```
root@ntp-VirtualBox:/home/ntp# netplan apply
root@ntp-VirtualBox:/home/ntp# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:40:56:d0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.21/24 brd 192.168.50.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe40:56d0/64 scope link tentative
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:5d:52:64 brd ff:ff:ff:ff:ff:ff
    inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute enp0s8
        valid_lft 86400sec preferred_lft 86400sec
    inet6 fe80::a00:27ff:fe5d:5264/64 scope link tentative
        valid_lft forever preferred_lft forever
root@ntp-VirtualBox:/home/ntp#
```

Le serveur SRV-NTP a été configuré avec une adresse IP statique 192.168.50.21 sur l'interface connectée au réseau interne, afin d'assurer son accessibilité permanente en tant que serveur DNS secondaire.

on installe le serveur dns:

- `sudo apt update`
- `sudo apt install bind9 bind9utils dnsutils -y`


```

ntp@ntp-VirtualBox:~$ sudo apt update
[sudo] Mot de passe de ntp :
Atteint :1 http://security.ubuntu.com/ubuntu noble-security InRelease
Atteint :2 http://archive.ubuntu.com/ubuntu noble InRelease
Atteint :3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Atteint :4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
289 paquets peuvent être mis à jour. Exécutez « apt list --upgradable » pour les voir.
ntp@ntp-VirtualBox:~$ sudo apt install bind9 bind9utils dnsutils -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les paquets supplémentaires suivants seront installés :
  bind9-dnsutils bind9-host bind9-libs bind9-utils
Paquets suggérés :
  bind-doc
Les NOUVEAUX paquets suivants seront installés :
  bind9 bind9-utils bind9utils dnsutils
Les paquets suivants seront mis à jour :
  bind9-dnsutils bind9-host bind9-libs
3 mis à jour, 4 nouvellement installés, 0 à enlever et 286 non mis à jour.
Il est nécessaire de prendre 1 887 ko dans les archives.
Après cette opération, 1 688 ko d'espace disque supplémentaires seront utilisés.

```

on modifie le conf.options:

- sudo nano /etc/bind/named.conf.options

```

GNU nano 7.2 /etc/bind/named.conf.options *
options {
    directory "/var/cache/bind";

    recursion yes;
    allow-recursion { 192.168.50.0/24; };

    listen-on { 127.0.0.1; 192.168.50.21; };

    dnssec-validation auto;

    listen-on-v6 { any; };
};

```

on modifie le conf.local:

- sudo nano /etc/bind/named.conf.local

```
ntp@ntp-VirtualBox: ~  
GNU nano 7.2 /etc/bind/named.conf.local *  
  
zone "zl-cyber.local" {  
    type slave;  
    masters { 192.168.50.20; };  
    file "/var/cache/bind/db.zl-cyber.local";  
};  
  
zone "50.168.192.in-addr.arpa" {  
    type slave;  
    masters { 192.168.50.20; };  
    file "/var/cache/bind/db.192.168.50";  
};  
  
//  
// Do any local configuration here  
//  
  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";
```

on autorise le transfert sur ldap:

- sudo nano /etc/bind/named.conf.local

```
GNU nano 7.2 /etc/bind/named.conf.local *  
zone "zl-cyber.local" {  
    type master;  
    file "/etc/bind/db.zl-cyber.local";  
    allow-transfer { 192.168.50.21; };  
};  
  
zone "50.168.192.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.192.168.50";  
    allow-transfer { 192.168.50.21; };  
};  
  
//  
// Do any local configuration here  
//  
  
// Consider adding the 1918 zones here, if they are not used in your  
// organization  
//include "/etc/bind/zones.rfc1918";
```

192.168.50.21 = IP de SRV-NTP (serveur esclave)

on teste:

- sudo named-checkconf
- sudo systemctl restart bind9

```
root@ldap-VirtualBox:/home/ldap# nano /etc/bind/named.conf.local
root@ldap-VirtualBox:/home/ldap# named-checkconf
root@ldap-VirtualBox:/home/ldap# sudo systemctl restart bind9
root@ldap-VirtualBox:/home/ldap#
```

sur SRV-NTP:

vérification:

- sudo named-checkconf
- sudo systemctl restart bind9
- ls /var/cache/bind

```
ntp@ntp-VirtualBox:~$ sudo named-checkconf
ntp@ntp-VirtualBox:~$ sudo systemctl restart bind9
ntp@ntp-VirtualBox:~$ ls /var/cache/bind
db.192.168.50 db.zl-cyber.local managed-keys.bind managed-keys.bind.jnl
ntp@ntp-VirtualBox:~$
```

```

ntp@ntp-VirtualBox:~$ dig @127.0.0.1 srv-ldap.zl-cyber.local

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @127.0.0.1 srv-ldap.zl-cyber.local
; (1 server found)
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 56211
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: cb639a3c3f63686001000000696a1ab9e8d8e04e516deff8 (good)
;; QUESTION SECTION:
;srv-ldap.zl-cyber.local.      IN      A

;; ANSWER SECTION:
srv-ldap.zl-cyber.local. 604800 IN      A      192.168.50.20

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Jan 16 12:02:17 CET 2026
;; MSG SIZE rcvd: 96

ntp@ntp-VirtualBox:~$ dig @127.0.0.1 -x 192.168.50.20

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> @127.0.0.1 -x 192.168.50.20
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 34151
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 47735774b4eae8601000000696a1abe0916034e7122de92 (good)
;; QUESTION SECTION:
;20.50.168.192.in-addr.arpa.  IN      PTR

;; ANSWER SECTION:
20.50.168.192.in-addr.arpa. 604800 IN      PTR      srv-ldap.zl-cyber.local.

;; Query time: 0 msec
;; SERVER: 127.0.0.1#53(127.0.0.1) (UDP)
;; WHEN: Fri Jan 16 12:02:22 CET 2026
;; MSG SIZE rcvd: 120

```

Les fichiers de zones ont été automatiquement créés sur le serveur secondaire dans le répertoire /var/cache/bind, confirmant la réussite du transfert de zones depuis le serveur maître.

sur la vm client:
id & mdp: srv1

```

2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:7c:55:b7 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 85799sec preferred_lft 85799sec
    inet6 fd17:625c:f037:2:77e7:51dc:856e:d20c/64 scope global temporary dynamic

        valid_lft 86179sec preferred_lft 14179sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7c:55b7/64 scope global dynamic mngtmpaddr

        valid_lft 86179sec preferred_lft 14179sec
    inet6 fe80::a00:27ff:fe7c:55b7/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
oup default qlen 1000
    link/ether 08:00:27:92:b0:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.50.10/24 brd 192.168.50.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe92:b01e/64 scope link
        valid_lft forever preferred_lft forever
root@srv1-VirtualBox:/home/srv1#

```

dans netplan client:

```

root@srv1-VirtualBox:/home/srv1
GNU nano 7.2 /etc/netplan/50-cloud-init.yaml *
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    enp0s3:
      dhcp4: true

    enp0s8:
      dhcp4: no
      addresses:
        - 192.168.50.10/24
      nameservers:
        addresses:
          - 192.168.50.20
          - 192.168.50.21
      search:
        - zl-cyber.local

```

on donne les droits:

- sudo chmod 600 /etc/netplan/*.yaml
- sudo netplan apply

```

root@srv1-VirtualBox:/etc/netplan# chmod 600 /etc/netplan/*.yaml
root@srv1-VirtualBox:/etc/netplan# netplan apply
root@srv1-VirtualBox:/etc/netplan#

```

On vérifie si le dns est bien actif:

```
root@srv1-VirtualBox: /etc/netplan

root@srv1-VirtualBox:/etc/netplan# resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (enp0s3)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.1.254
    DNS Servers: 192.168.1.254

Link 3 (enp0s8)
    Current Scopes: DNS
    Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.50.21
    DNS Servers: 192.168.50.20 192.168.50.21
    DNS Domain: zl-cyber.local
root@srv1-VirtualBox:/etc/netplan#
```

```
srv1@srv1-VirtualBox:~$ sudo su
root@srv1-VirtualBox:/home/srv1# ip neigh flush all
root@srv1-VirtualBox:/home/srv1# ping 192.168.50.20
PING 192.168.50.20 (192.168.50.20) 56(84) bytes of data.
64 bytes from 192.168.50.20: icmp_seq=1 ttl=64 time=52.4 ms
64 bytes from 192.168.50.20: icmp_seq=2 ttl=64 time=9.64 ms
^C
--- 192.168.50.20 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 9.639/31.022/52.406/21.383 ms
root@srv1-VirtualBox:/home/srv1#
```

```
root@srv1-VirtualBox:/home/srv1# ping srv-ldap.zl-cyber.local
PING srv-ldap.zl-cyber.local (192.168.50.20) 56(84) bytes of data.
64 bytes from srv-ldap.zl-cyber.local (192.168.50.20): icmp_seq=1 ttl=64 time=1.69 ms
64 bytes from srv-ldap.zl-cyber.local (192.168.50.20): icmp_seq=2 ttl=64 time=25.3 ms
64 bytes from srv-ldap.zl-cyber.local (192.168.50.20): icmp_seq=3 ttl=64 time=2.16 ms
64 bytes from srv-ldap.zl-cyber.local (192.168.50.20): icmp_seq=4 ttl=64 time=10.6 ms
^C
--- srv-ldap.zl-cyber.local ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3012ms
rtt min/avg/max/mdev = 1.694/9.936/25.253/9.532 ms
root@srv1-VirtualBox:/home/srv1#
```

le ping [srv-ntp.zl-cyber.local](#) puisque l' on a pas ajouter l' ip sur le dns 1 on redémarre les 2 dns par la suite .

sur ldap:

- sudo nano /etc/bind/db.zl-cyber.local

```
ldap@ldap-VirtualBox: ~  
GNU nano 7.2 /etc/bind/db.zl-cyber.local *  
$TTL      604800  
@          IN      SOA      srv-ldap.zl-cyber.local. admin.zl-cyber.local. (  
                2026011601      ; Serial  
                604800      ; Refresh  
                86400      ; Retry  
                2419200      ; Expire  
                604800 )      ; Negative Cache TTL  
;  
@          IN      NS       srv-ldap.zl-cyber.local.  
srv-ldap   IN      A        192.168.50.20  
srv-ntp    IN      A        192.168.50.21
```

puis:

- sudo named-checkzone zl-cyber.local /etc/bind/db.zl-cyber.local
- sudo systemctl restart bind9

on redémarre le dns 1:

```
ldap@ldap-VirtualBox:~$ sudo named-checkzone zl-cyber.local /etc/bind/db.zl-cyber.local  
zone zl-cyber.local/IN: loaded serial 2026011601  
OK  
ldap@ldap-VirtualBox:~$ sudo systemctl restart bind9  
Warning: The unit file, source configuration file or drop-ins of bind9.service changed on disk.  
. Run 'systemctl daemon-reload' to reload units.  
ldap@ldap-VirtualBox:~$ sudo systemctl daemon-reload  
ldap@ldap-VirtualBox:~$ sudo systemctl restart bind9  
ldap@ldap-VirtualBox:~$
```

sur ntp:

on redémarre le dns 2 aussi

```
ntp@ntp-VirtualBox:~$ sudo systemctl restart bind9  
[sudo] Mot de passe de ntp :  
Warning: The unit file, source configuration file or drop-ins of bind9.service changed on disk.  
Run 'systemctl daemon-reload' to reload units.  
ntp@ntp-VirtualBox:~$ sudo systemctl daemon-reload  
ntp@ntp-VirtualBox:~$ sudo systemctl restart bind9  
ntp@ntp-VirtualBox:~$
```

sur la client:


```
root@srv1-VirtualBox: /home/srv1
root@srv1-VirtualBox:/home/srv1# ping srv-ntp.zl-cyber.local
PING srv-ntp.zl-cyber.local (192.168.50.21) 56(84) bytes of data.
64 bytes from 192.168.50.21: icmp_seq=1 ttl=64 time=7.86 ms
64 bytes from 192.168.50.21: icmp_seq=2 ttl=64 time=8.03 ms
^C
--- srv-ntp.zl-cyber.local ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1005ms
rtt min/avg/max/mdev = 7.856/7.941/8.026/0.085 ms
root@srv1-VirtualBox:/home/srv1#
```

éteindre la vm du dns 1:

ex5:

on installe la commande

- sudo apt install dnsutils -y

```
root@srv1-VirtualBox:/home/srv1# apt install dnsutils -y
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
Les NOUVEAUX paquets suivants seront installés :
  dnsutils
0 mis à jour, 1 nouvellement installés, 0 à enlever et 147 non mis à jour.
Il est nécessaire de prendre 3 684 o dans les archives.
Après cette opération, 31,7 ko d'espace disque supplémentaires seront utilisés.
Réception de :1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 dnsutils al
l 1:9.18.39-0ubuntu0.24.04.2 [3 684 B]
3 684 o réceptionnés en 0s (22,1 ko/s)
Sélection du paquet dnsutils précédemment désélectionné.
(Lecture de la base de données... 153659 fichiers et répertoires déjà installés.)
Préparation du dépaquetage de .../dnsutils_1%3a9.18.39-0ubuntu0.24.04.2_all.deb ...
Dépaquetage de dnsutils (1:9.18.39-0ubuntu0.24.04.2) ...
Paramétrage de dnsutils (1:9.18.39-0ubuntu0.24.04.2) ...
root@srv1-VirtualBox:/home/srv1#
```

```
root@srv1-VirtualBox: /home/srv1
root@srv1-VirtualBox:/home/srv1# nslookup srv-ldap.zl-cyber.local
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   srv-ldap.zl-cyber.local
Address: 192.168.50.20
;; communications error to 127.0.0.53#53: timed out

root@srv1-VirtualBox:/home/srv1# resolvectl flush-caches
root@srv1-VirtualBox:/home/srv1# resolvectl statistics
Transactions
                Current Transactions:  0
                Total Transactions: 58

Cache
                Current Cache Size:  1
                Cache Hits: 8
                Cache Misses: 51

Failure Transactions
                Total Timeouts: 1
                Total Timeouts (Stale Data Served): 0
                Total Failure Responses: 0
                Total Failure Responses (Stale Data Served): 0

DNSSEC Verdicts
                Secure: 0
                Insecure: 0
                Bogus: 0
                Indeterminate: 0
root@srv1-VirtualBox:/home/srv1#
```

requête vers le dns 1:

```
root@srv1-VirtualBox:/home/srv1# nslookup srv-ldap.zl-cyber.local
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   srv-ldap.zl-cyber.local
Address: 192.168.50.20

root@srv1-VirtualBox:/home/srv1#
```

forcer la requête vers le serveur dns 2:

```
root@srv1-VirtualBox:/home/srv1# nslookup srv-ldap.zl-cyber.local 192.168.50.21
Server:          192.168.50.21
Address:         192.168.50.21#53

Name:   srv-ldap.zl-cyber.local
Address: 192.168.50.20

root@srv1-VirtualBox:/home/srv1#
```

Le mécanisme de résolution DNS du client permet une bascule automatique vers le serveur secondaire lorsque le serveur maître devient indisponible. Le test par requête directe vers le serveur secondaire confirme la redondance et la haute disponibilité du service DNS.